

HARDEEP GUMBER, PhD

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PROFESSIONAL SUMMARY

A PhD Research Scientist and 5+ years of experience generating and leveraging data to uncover insights in fast-paced R&D environments. Skilled in experimental design, running lab and clinical studies, NGS data analysis, statistical analysis, and communicating complex findings to both technical and non-technical stakeholders. Proven ability to drive innovation, collaborate cross-functionally, and adapt quickly—ideal for dynamic, data-driven organizations.

CORE STRENGTHS

- Experimental Design & Hypothesis Testing
- Data generation, analysis and interpretation
- Clinical and lab research
- Insight Discovery & Reporting
- Cross-functional Communication
- Team player and leader

TECHNICAL SKILLS

Python (Pandas, NumPy, Seaborn), R (tidyverse, ggplot2), SQL, Excel, Tableau, Git, Jupyter Notebooks, Microscopy, Mass spectrometry, proteomics, NGS, 16s rRNA gene sequencing, gene cloning, interpretation of multi-omics data, PCR, qPCR, Western Blot, gel electrophoresis, co-immunoprecipitation, multiplex ELISA assays, sample preparation, tissue sections, confocal microscopy, SEM.

PROFESSIONAL EXPERIENCE

Senior Research Scientist

Colgate-Palmolive, Piscataway, NJ

2019 – Present

- Conducted dental clinical studies to support discovery in oral care.
- Compiled and analyzed data to draw biological insights and identify modes of actions of products.
- Led initiatives to derive mode of action of products, translating complex data into actionable narratives for R&D and marketing teams.
- Developed reproducible methods for experimental testing and written SOPs.
- Presented key findings at international conferences (e.g., IADR), contributing to organization's innovation pipeline.
- Collaborated with interdisciplinary teams (product development, regulatory, marketing) to ensure insights were timely and impactful.
- Published in peer-reviewed scientific journals; co-inventor on a patent pending.

EDUCATION

- **PhD** Cell and Molecular Biology (2019) Florida State University, Tallahassee, Florida, USA.

- **MS** Biotechnology (2011) Punjab Agricultural University, Ludhiana, Punjab, India.
- **BS** Biology and Chemistry (2008) Punjab University, Chandigarh, India.
- **Data Analytics Bootcamp** (2025) University of California Berkley Extension, Berkley, California, USA.

PEER-REVIEWED PUBLICATIONS

- Gumber HK, Louyakis AS, Sarma T, Fabijanic KI, Paul R, Mellenbruch K, Kilpatrick-Liverman L (2022). "Effect of a Stannous Fluoride Dentifrice on Biofilm Composition, Gene Expression and Biomechanical Properties". *Microorganisms* 10(9):1691.
- Gumber HK, McKenna JF, Estrada AL, Tolmie AF, Graumann K and Bass HW (2019). "Identification and characterization of genes encoding the nuclear envelope LINC complex in the monocot species *Zea mays*". *J Cell Science* 132: jcs221390.
- Gumber HK, McKenna JF, Tolmie AF, Jalovec AM, Kartick AC, Graumann K and Bass HW (2019). "MLKS2 is an ARM domain and F-actin-associated KASH protein that functions in stomatal complex development and meiotic chromosome segregation". *Nucleus* 10:144-166.
- Murphy SP, Gumber HK, Mao YY, and Bass HW (2014). "A dynamic meiotic SUN belt includes the zygotene-stage telomere bouquet and is disrupted in chromosome segregation mutants of maize (*Zea mays* L.). *Front Plant Sci.* 5:314.
- Bass HW, Wear EE, Lee T-J, Hoffman GG, Gumber HK, Allen GC, Thompson WF, Hanley-Bowdoin L (2014). "A maize root tip system to study DNA replication programmes in somatic and endocycling nuclei during plant development" *J Exp Botany* 65:2747-2756.
- McKenna JF*, Gumber HK*, Turpin ZM, Jalovec AM, Kartick AC, Graumann K and Bass HW (2021) "Maize (*Zea mays* L.) nucleoskeletal proteins regulate nuclear envelope remodeling and function in stomatal complex development and pollen viability". *Frontiers in Plant Science* 12:645218. (*Co-first authors)